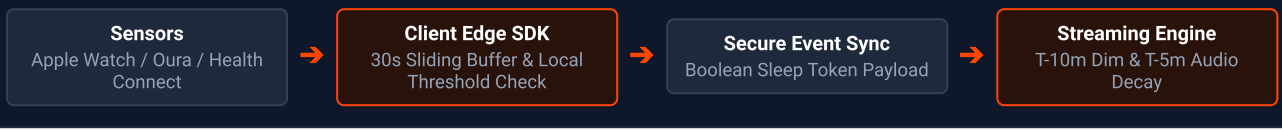
NETFLIX AFTER HOURS
PATH A: COMPLETE TECHNICAL ARCHITECTURE & DIAGRAMS

Document Version: 2.0-FINAL
Engineering System Specs

PATH A1: REAL-TIME BIOMETRIC SLEEP TELEMETRY PIPELINE ARCHITECTURE

The event-driven telemetry pipeline ingest biometric data from wearables, processes sliding windows locally on client edge SDKs, and dispatches sleep state tokens to trigger playback fade curves.

FIGURE 1.1: WEARABLE INGESTION & PLAYBACK CONTROLLER FLOW



Platform	Ingest Mechanism	Biometric Marker	Trigger Threshold
Apple HealthKit	`HKCategoryTypeIdentifierSleepAnalysis`	HRV drop + Motion quiescence	`HKCategoryValueSleepAnalysisAsleep` / N1 Stage ≥ 45s
Health Connect	`SleepSessionRecord` Listener	Resting HR drop + Accelerometer	`STAGE_N1` or `STAGE_N2` event payload
Oura API v2	Webhook / REST Polling (1-min)	Skin Temp drop + RHR stabilization	`sleep_state: "light"` or `"deep"`

BIOMETRIC TELEMETRY JSON INGESTION PAYLOAD

```
{ "event_header": { "event_id": "evt_98320491-a3bf", "timestamp_utc": "2027-04-18T03:42:15.120Z" },
"user_context": { "account_id": "usr_8839201", "profile_id": "prf_4401", "tier": "premium" },
"biometric_telemetry": { "provider": "apple_healthkit", "sleep_stage_detected": "N1_LIGHT_SLEEP",
"confidence_score": 0.94 }, "playback_state": { "content_id": "80057281", "episode_id": "80192011",
"current_timestamp_ms": 860400 }, "action_triggered": "INITIATE_DECAY_FADE_SEQUENCE" }
```

PRIVACY & SECURITY COMPLIANCE SAFEGUARDS

- **Zero Cloud Health Storage:** Raw metrics remain locally on client edge; only boolean sleep tokens transfer to servers.
- **Explicit OS Opt-in:** Requires explicit consent: *"Allow Netflix to access Sleep Analysis data to automatically pause streaming."*

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PATH A2: CLIENT-SIDE AUDIO DSP ARCHITECTURE

Document Version: 2.0-FINAL
Audio DSP Pipeline

PATH A2: AUDIO DSP PIPELINE & SPATIAL LAYERING SCHEMATICS

Real-time client-side DSP pipeline structure operating between stream demuxing and hardware audio output buffers.

FIGURE 2.1: MULTI-STAGE AUDIO DSP & SPATIAL LAYERING FLOW



Stage 1 — High-Cut Low-Pass Filter: 2nd-order Butterworth IIR filter (12 dB/octave). Cutoff scales from 20 kHz down to 1.8 kHz over 300 seconds to preserve dialogue while stripping transient harshness.

Stage 2 — Dynamic Range Compression (DRC): Scales threshold from -12 dBFS to -28 dBFS with ratio increasing from $2:1$ to $8:1$ to lock dialogue clarity during volume drop.

LOGARITHMIC PERCEPTUAL GAIN DECAY PROFILE

```
Gain(t) = Gain_start * 10^((Target_dB * t) / (20 * T_fade)) T-5m (100%): 0 dBFS | fc: 20.0 kHz (Full
Spectrum) T-3m ( 60%): -12 dBFS | fc: 8.0 kHz (Transients Attenuated) T-1m ( 20%): -24 dBFS | fc: 2.2 kHz
(Voice Intelligibility Retained) T-0m ( 0%): -48 dBFS | Mute & Shutdown
```

SPATIAL AUDIO & BINAURAL SLEEP LAYERING (PREMIUM TIER)

- **Delta Wave Generation:** Dual sine generators emitting 100.0 Hz (Left) and 103.0 Hz (Right) to create a 3.0 Hz Delta brainwave beat.
- **Fixed-Head Spatial Lock:** Head-tracking displacement is disabled during bedtime mode to prevent spatial pan shifts when turning on a pillow.

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PATH A3: OFFLINE-FIRST LOCAL & CLOUD SYNC ARCHITECTURE

Document Version: 2.0-FINAL
Sync State Engine

PATH A3: OFFLINE-FIRST SYNC & CRDT CONFLICT RESOLUTION ENGINE

Distributed CRDT state machine diagram resolving local decay bookmarks against remote cloud progress markers.



Platform	Local Engine	Encryption & Security	Write Guarantee
iOS / iPadOS	CoreData + SQLite	`NSFileProtectionCompleteUnlessOpen`	< 5 ms sync write
Android / TV	Room Persistence	EncryptedSharedPreferences + SQLCipher	< 5 ms async coroutine
Smart TV OS	IndexedDB / Native DB	OS Secure Storage Sandbox	< 12 ms transactional

CONFLICT RESOLUTION MATRIX

Scenario	Local State	Cloud State	Resolved Outcome
Offline Flight	Offline Decay BM: 14:20	No Cloud Update	**Local BM Prevails** (Renders recovery card offline)
Dual Device	TV Decay BM: 14:20	Phone Progress BM: 42:10	**Decay BM 14:20 Wins** (Protects from spoilers)
Manual Override	Decay BM: 14:20	Manual Scrub: 45:00	**Manual Scrub Wins** (Active input overrides passive)

CLIENT SYNC DISPATCH PAYLOAD

```
{ "sync_header": { "client_tx_id": "tx_99812039-44ab", "device_id": "dev_ios_mac_8839" },
"pending_bookmarks": [{ "bookmark_id": "bm_8839201_80192011", "account_id": "usr_8839201", "profile_id":
"prf_4401", "decay_timestamp_ms": 860400, "trigger_type": "wearable_auto", "vector_clock": {
"device_node": "dev_ios_mac_8839", "counter": 4 } }] }
```